

Reading Base-Ten Models: Hundreds, Tens, and Ones

Answer Key

Grade 2–3 Place Value

Quick Answers

1. 243	3. 572	5. 60	7. —
2. 306	4. 180	6. 335	8. —

1. 2 flats, 4 rods, 3 units. Flats: $2 \times 100 = 200$. Rods: $4 \times 10 = 40$. Units: $3 \times 1 = 3$. Read it as “2 hundreds, 4 tens, 3 ones”, which is $200 + 40 + 3$. Each digit sits in the place its blocks name: 243.

2. 3 flats, 0 rods, 6 units. Flats: $3 \times 100 = 300$. There are no rods, so the tens place is empty — but it still needs a digit to hold the ones out of the tens column, and that digit is 0. Units: 6. So $300 + 0 + 6 =$ 306. (Writing “36” is the classic slip: it would mean 3 tens and 6 ones.)

3. 5 flats, 7 rods, 2 units. Expanded form writes each place’s value on its own:

$$5 \times 100 = 500, \quad 7 \times 10 = 70, \quad 2 \times 1 = 2$$

so the expanded form is $500 + 70 + 2$, and adding the parts gives 572.

4. 1 flat, 8 rods, 0 units. $1 \times 100 = 100$, $8 \times 10 = 80$, $0 \times 1 = 0$, so the expanded form is $100 + 80 + 0$. Adding gives 180. The 0 in the ones place keeps the 8 in the tens place — “18” would be a different number.

5. 4 flats, 6 rods, 7 units — value of the rods. There are 6 rods, and one rod is worth ten. Six rods are therefore worth $6 \times 10 =$ 60. The whole model reads 467, and the 6 in it stands for sixty, not six: its place tells its value.

6. 2 flats, 13 rods, 5 units — trade first. Thirteen rods is more than nine, so trade ten of them for one flat:

$$13 \text{ rods} = 1 \text{ flat} + 3 \text{ rods}$$

Now count again: $2 + 1 = 3$ flats, 3 rods, 5 units, so $300 + 30 + 5 =$ 335. Check without trading: $200 + 130 + 5 = 335$ — same number, as it must be.

7. Compare 3 flats, 4 rods, 2 units with 3 flats, 2 rods, 8 units. Model A = $300 + 40 + 2 = 342$. Model B = $300 + 20 + 8 = 328$. Compare from the biggest place down: the hundreds tie at 3, so look at the tens. 4 tens beats 2 tens, and that settles it — the ones never get a vote. So Model A > Model B.

8. Challenge: 4 flats, 2 rods, 6 units versus 3 flats, 12 rods, 9 units. Model A = $400 + 20 + 6 = 426$. Model B has twelve rods, so trade ten rods for one flat: $3 + 1 = 4$ flats, 2 rods, 9 units, giving $400 + 20 + 9 = 429$. Now both have 4 hundreds and 2 tens, so the ones decide: $6 < 9$. Model A Model B. Rosa counted flats before trading; after the trade both models have the same number of hundreds, so “more flats on the table” was never the right test.